

PES UNIVERSITY • DEPT. OF CSE • LAB 1

Academic Elective Bidding & Allocation System

Student Name: Shashank D | **SRN:** PES1UG24CS433 | **Problem Statement #05:**
Requirements Engineering & UML Use-Case Modelling

1. Requirements Table

2. Main Use Case Diagram

3. Main Flow Specification

4. Alternate Flow Diagram & Specs

5. Exception Flow Document

1. Complete Requirements Table

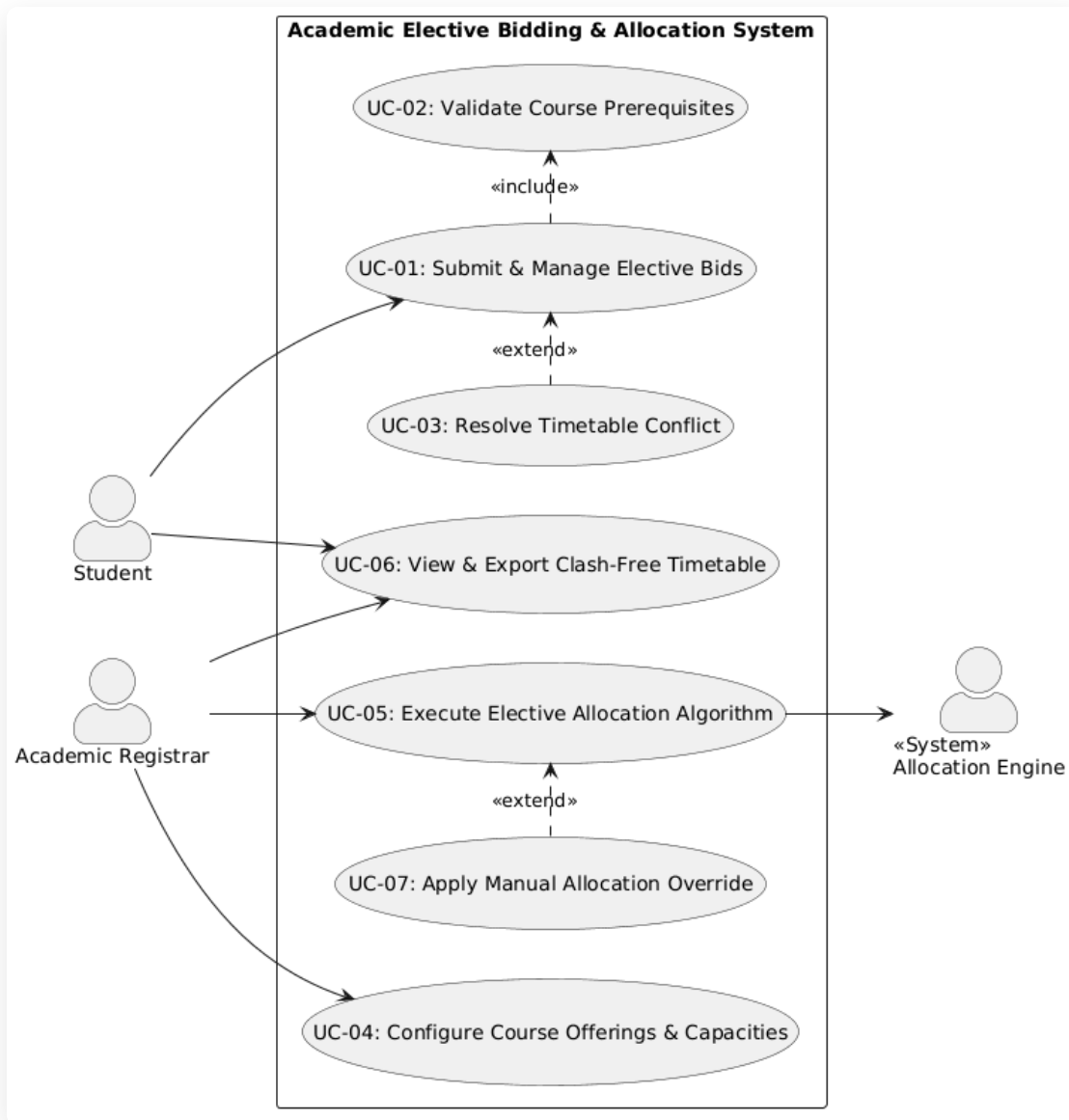
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow students to distribute 100 bidding credits across their ranked elective preferences and validate prerequisite course completions.	High	Pass: Credits deduct accurately (sum == 100) and prerequisites validated. Fail: Total allocated credits != 100 or unmet prerequisites bypassed.	Ensures fair bid distribution aligned with student priority and academic eligibility.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-002	Functional	The system shall allow the Academic Registrar to configure elective course offerings, section capacities (min/max seats), and designated timetable slot constraints.	High	Pass: Course records saved with designated capacities and slot times. Fail: Duplicate slot assignments without multi-section flag or invalid seat caps.	Establishes the boundary conditions and constraints required by the automated allocation solver.
FR-003	Functional	The system shall execute the multi-criteria optimization solver to allocate courses based on bid weightings, CGPA tie-breakers, and room/slot collision avoidance rules.	High	Pass: All eligible bids evaluated; seats filled up to max limit with 0 timetable clashes. Fail: Student assigned to overlapping lecture slots or oversubscribed course.	Core algorithmic automation eliminating manual clash resolution and ensuring objective allocation.
FR-004	Functional	The system shall allow students to view real-time provisional seat demand percentages and adjust/reallocate their submitted bidding points	Medium	Pass: Bid updates reflected instantly in student profile and draft pool before lock time. Fail:	Provides transparency on course popularity, allowing students to strategize point

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		prior to the deadline window closure.		Modifications permitted after deadline timestamp.	allocation dynamically.
FR-005	Functional	The system shall publish final allocation results, generate individualized clash-free timetables for students, and export master enrollment rosters for the Academic Registrar.	Medium	Pass: Downloadable PDF/ICS schedule produced with 0 slot clashes; Registrar roster matches allocated student IDs. Fail: Incomplete roster export or unassigned enrolled courses.	Closes the registration lifecycle with actionable student schedules and official registrar records.
NFR-001	Non-Functional (Performance)	The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds.	High	Pass: Benchmarking tests confirm execution time < 30 seconds for 5,000 concurrent student bids under simulated peak load. Fail: Solver execution exceeds 30 seconds or server runs out of memory.	Prevents server bottlenecks and ensures rapid turnaround between bidding closure and timetable publishing.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-002	Non-Functional <i>(Security)</i>	The system shall enforce Role-Based Access Control (RBAC) and maintain an immutable audit trail of all bid submissions, modifications, and administrative overrides.	High	Pass: Unauthorized role access blocked (403 Forbidden); 100% of bid modification events logged with timestamp, user ID, and prior state. Fail: Unauthenticated bid injection or unlogged state change.	Guarantees academic integrity, prevents tampering, and ensures auditability during student grievance redressals.

2. Main UML Use-Case Diagram



3. Use-Case Flow Specification (Main Flow)

Use Case: UC-01 — Submit Elective Bids

Primary Actor: Student | **Secondary Actor:** Academic Database / Prerequisite Service | **Priority:** High

Preconditions:

- Student is authenticated and has active enrollment status.

- Elective bidding window is active/open.
- Student has exactly 100 unspent bidding points available for the semester.

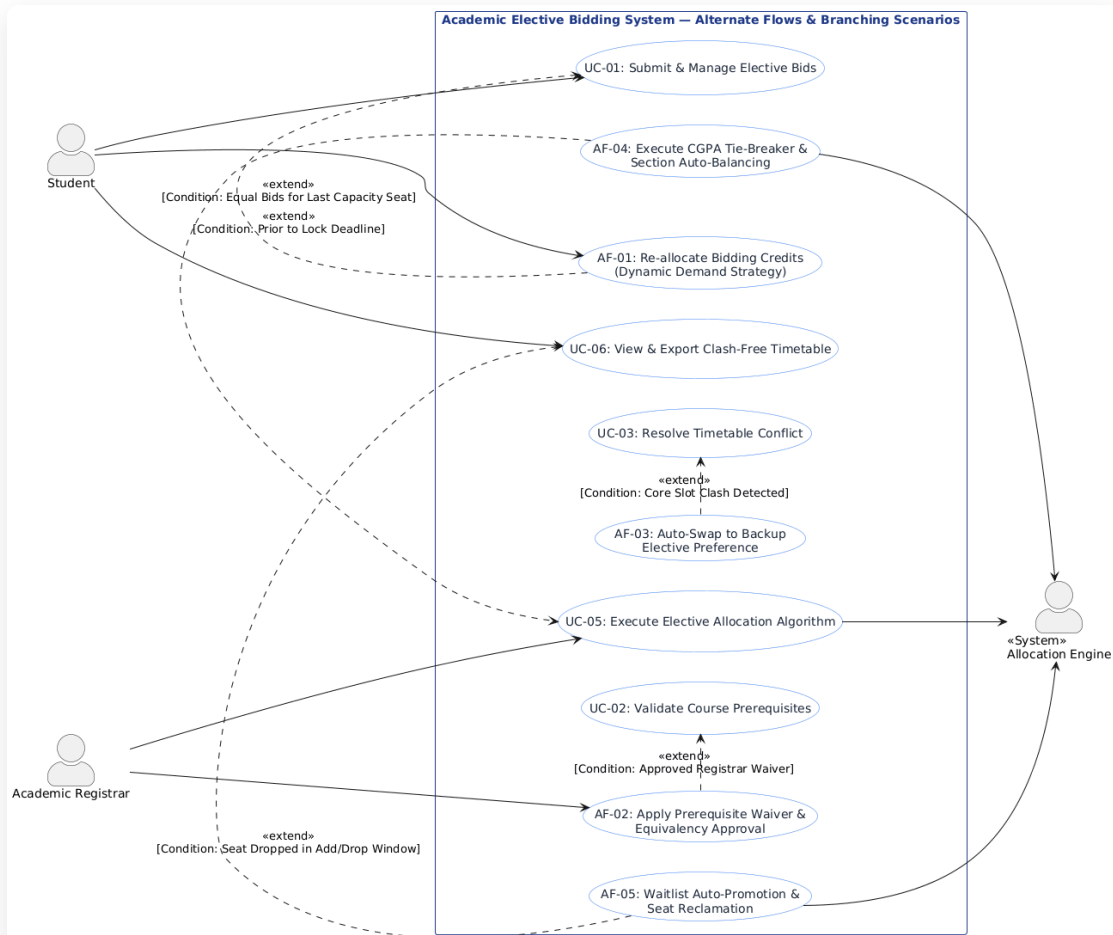
Postconditions:

- Student's elective bids and credit allocations are persistently stored in the database.
- Total credits used are marked as 100.
- An immutable audit log entry and confirmation receipt are generated.

Main Success Scenario

1. Student logs into the portal and navigates to the "Elective Course Bidding" section.
2. System displays the list of offered elective courses, descriptions, available seats, faculty, and schedule slot timings.
3. Student selects desired elective courses and distributes exactly 100 bidding points among them based on priority.
4. Student submits the bidding form by clicking "Submit Bids".
5. System executes «include» **UC-02: Validate Course Prerequisites** to check completed credits against prerequisites.
6. System validates that the sum of allocated bidding points is exactly equal to 100.
7. System confirms that no two selected electives share mutually exclusive required core slots.
8. System commits the bid records into the active allocation pool and generates a transaction timestamp.
9. System displays confirmation message and locked preference summary.
10. Use Case ends successfully.

4. Alternate Flow Use-Case Diagram & Specifications



Alternate Flow Specifications

- **AF-01: Re-allocate Bidding Credits (Dynamic Demand Strategy - «extend» UC-01)**

Condition: Prior to Bidding Deadline Closure.

Flow: Student reviews dynamic seat popularity metrics; adjusts points across draft electives to maximize allocation likelihood; saves updated 100-credit distribution.

- **AF-02: Prerequisite Waiver & Equivalency Approval («extend» UC-02)**

Condition: Registrar Waiver Granted for Course Prerequisites.

Flow: Student lacks standard prerequisite but presents signed registrar waiver token; system bypasses standard check and approves bid submission.

- **AF-03: Auto-Swap to Backup Elective Preference («extend» UC-03)**
Condition: Primary Elective Timetable Clash Detected.
Flow: When top-ranked preference conflicts with core slot, engine automatically shifts evaluation to pre-ranked secondary backup course without terminating submission.
- **AF-04: Execute CGPA Tie-Breaker & Section Auto-Balancing («extend» UC-05)**
Condition: Equal Bid Points for Final Capacity Seat.
Flow: Solver breaks point ties using Cumulative GPA and prerequisite grade index, assigning tied student to parallel open section slot.
- **AF-05: Waitlist Auto-Promotion & Seat Reclamation («extend» UC-06)**
Condition: Seat Vacated during Add/Drop Window.
Flow: Engine reclaims vacated seat and automatically promotes highest waitlisted student, updating timetable and dispatching notification.

5. Exception Flow Specification Document

Exception Summary Matrix

Exception ID	Target UC	Exception Title	Error Category	Severity	Detection & System Action
EF-01	UC-01	Credit Sum Mismatch (\$\neq 100\$)	Data Validation	High	Halt commit; highlight fields; display dynamic point counter; prompt readjustment.

Exception ID	Target UC	Exception Title	Error Category	Severity	Detection & System Action
EF-02	UC-01 / UC-02	Unmet Prerequisite Check Failure	Domain Rule	High	Reject line-item bid; flag transcript gap; prompt course swap or waiver submission.
EF-03	UC-01 / UC-03	Unresolvable Slot Collision	Hard Constraint	High	Flag schedule clash with core slots; reject invalid matrix; prompt non-conflicting choice.
EF-04	UC-01	Deadline Expiry Lockout	Temporal / State	Critical	Abort transaction; enforce READ_ONLY lock; preserve prior pre-deadline submission state.
EF-05	UC-05	Solver Convergence / Timeout	Performance	Critical	Watchdog kills process at 30s limit; roll back DB checkpoint; alert Registrar for relaxation.

Exception ID	Target UC	Exception Title	Error Category	Severity	Detection & System Action
EF-06	UC-01 / UC-05	DB Concurrency Deadlock	Infrastructure	High	Catch DB lock; execute idempotent exponential retries; enqueue bid request if load spikes.
EF-07	UC-07	Unauthorized Override Attempt	Security / RBAC	Critical	Deny access (HTTP 403); write immutable security audit log entry; flag origin IP.

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Shashank D (PES1UG24CS433) — Software Engineering Lab 1 Complete Deliverables